CHAPTER 5 INTEGRALS

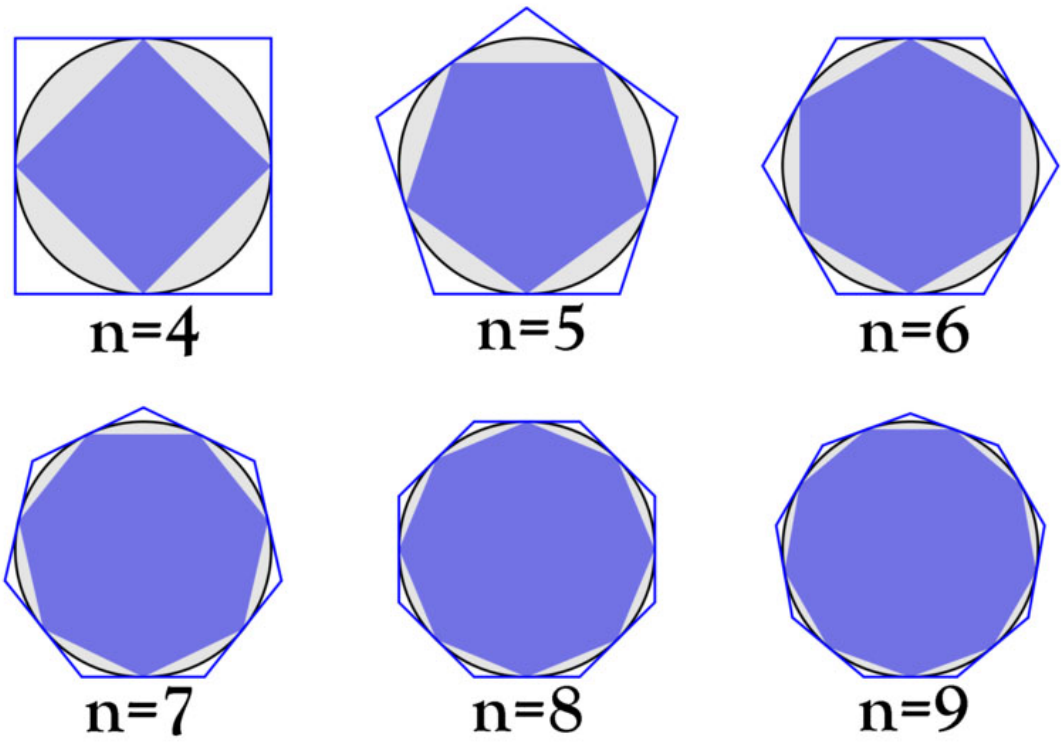
5.1 Areas and Distances

- How do we find the area of a circle?

  Over 2000 years ago, Archimedes, a famous Greek mathematician, provided the key to a solution.

  Consider a sequence of **inscribed polygons** that approximate the circle with greater and greater accuracy.

  Archimedes also considered a sequence of **circumscribed polygons** that approximate the circle with greater and greater accuracy.



Let $A(P_n)$ be the area of the polygon with n sides.

$$A(\text{circle}) = \lim_{n\to\infty} A(P_n)$$

- **Σ Notation**

$$\sum_{i=m}^{n} a_i = a_m + a_{m+1} + \ldots + a_{n-1} + a_n$$

Example 5.1. Write out the following sums. Simplify whenever possible.

(a) $\sum_{i=1}^{4} i^2 =$

(b) $\sum_{i=3}^{n} i =$

HW (c) $\sum_{j=0}^{5} 2^j =$

HW (d) $\sum_{k=1}^{n} \frac{1}{k} =$

HW (e) $\sum_{i=1}^{3} \frac{i-1}{i^2+3} =$

(f) $\sum_{i=1}^{4} 2 =$

- **Properties of Σ**

- $\sum_{i=1}^{n} ca_i = c\sum_{i=1}^{n} a_i$
- $\sum_{i=1}^{n} (a_i \pm b_i) = \sum_{i=1}^{n} a_i \pm \sum_{i=1}^{n} b_i$

Example 5.2. Suppose that $\sum_{i=1}^{100} a_i = 60$, $\sum_{i=1}^{100} b_i = 11$, find

$$\sum_{i=1}^{100} (2a_i - 3b_i + 4)$$

$$\sum_{i=1}^{100} (2a_i - 3b_i + 4) = \sum_{i=1}^{100} 2a_i - \sum_{i=1}^{100} 3b_i + \sum_{i=1}^{100} 4$$

$$= 2\sum_{i=1}^{100} a_i - 3\sum_{i=1}^{100} b_i + 4\sum_{i=1}^{100} 1$$

$$= 2(60) - 3(11) + 4(100) = 487$$

Example 5.3. Show that $\sum_{i=1}^{n} (a_{i+1} - a_i) = a_{n+1} - a_1$. Hence, find $\sum_{i=1}^{n} \left[(i+1)^2 - i^2\right]$.

- **Some Special Sum Formulas**

1. $\sum_{i=1}^{n} 1 = 1 + 1 + 1 + \cdots + 1 = n$
2. $\sum_{i=1}^{n} i = 1 + 2 + 3 + \cdots + n = \frac{n(n+1)}{2}$
3. $\sum_{i=1}^{n} i^2 = 1^2 + 2^2 + 3^2 + \cdots + n^2 = \frac{n(n+1)(2n+1)}{6}$
4. $\sum_{i=1}^{n} i^3 = 1^3 + 2^3 + 3^3 + \cdots + n^3 = \left[\frac{n(n+1)}{2}\right]^2$
5. $\sum_{i=1}^{n} i^4 = 1^4 + 2^4 + 3^4 + \cdots + n^4 = \frac{n(n+1)(2n+1)(3n^2+3n-1)}{30}$

Example 5.4. Find a formula for

$$R_n = \frac{1}{n}\left(\frac{1}{n}\right)^2 + \frac{1}{n}\left(\frac{2}{n}\right)^2 + \frac{1}{n}\left(\frac{3}{n}\right)^2 + \cdots + \frac{1}{n}\left(\frac{n}{n}\right)^2$$

.

$$= \frac{1}{n^3}\left[1^2 + 2^2 + 3^2 + \cdots + n^2\right]$$

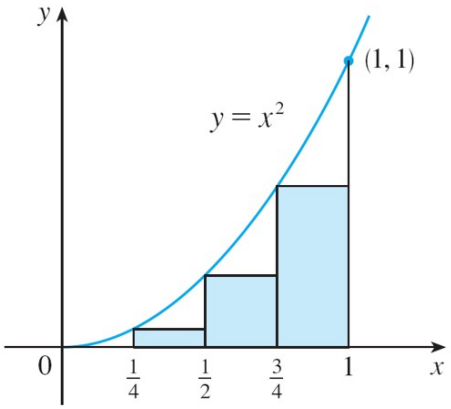
$$= \frac{1}{n^3}\sum_{i=1}^{n} i^2 = \frac{1}{n^3}\left[\frac{n(n+1)(2n+1)}{6}\right]$$

$$= \frac{(n+1)(2n+1)}{6n^2}$$

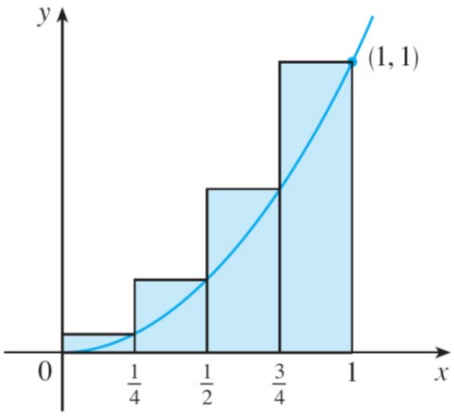
$$= \frac{1}{6}\cdot\frac{n+1}{n}\cdot\frac{2n+1}{n} = \frac{1}{6}\left(1+\frac{1}{n}\right)\left(2+\frac{1}{n}\right)$$

- **Areas**

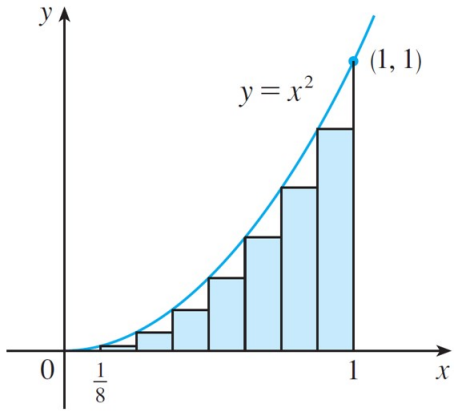
Example 5.5. Estimate the area of the region bounded by the curve $y = x^2$, the line $x = 1$ and the x-axis using four rectangles and left endpoints.



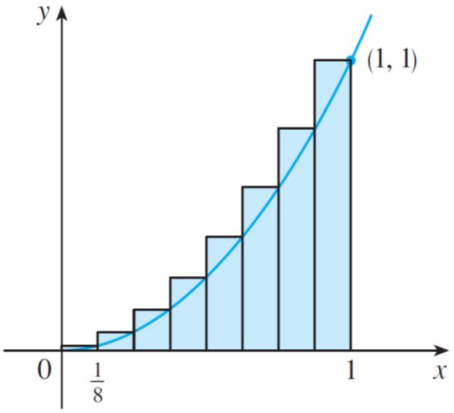
Example 5.6. Estimate the area of the region bounded by the curve $y = x^2$, the line $x = 1$ and the x-axis using four rectangles and right endpoints.



Example 5.7. Estimate the area of the region bounded by the curve $y = x^2$, the line $x = 1$ and the x-axis using eight rectangles and left endpoints.

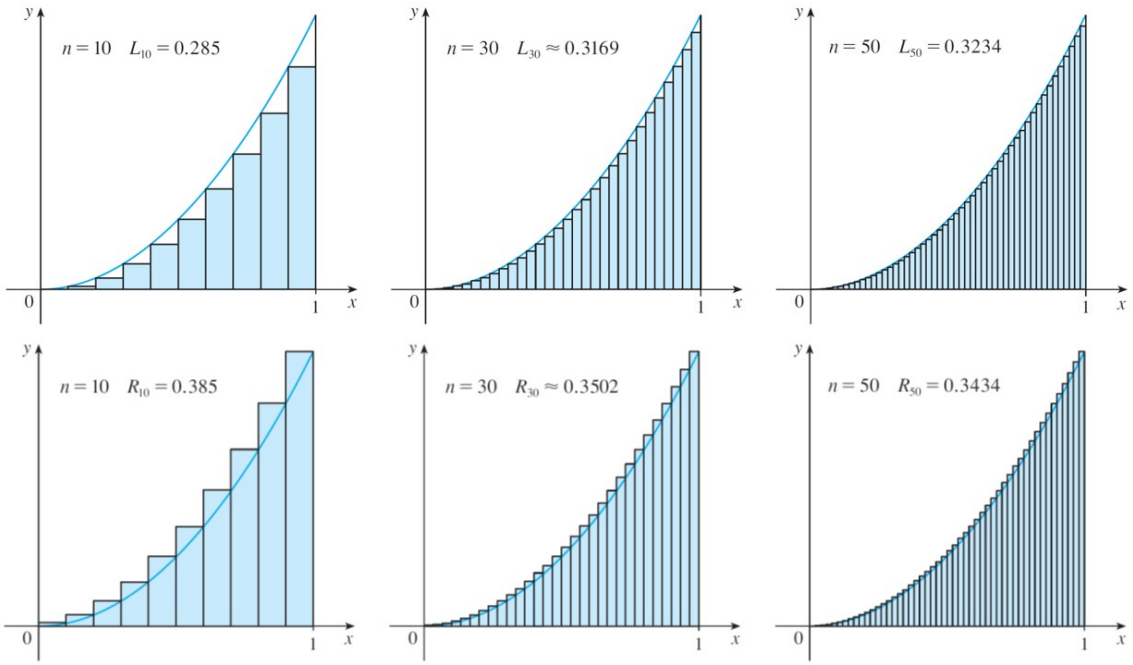


Example 5.8. Estimate the area of the region bounded by the curve $y = x^2$, the line $x = 1$ and the x-axis using eight rectangles and right endpoints.



Example 5.9. Find the area of the region bounded by the curve $y = x^2$, the line $x = 1$ and the x-axis using n rectangles and right endpoints with $n \to \infty$.

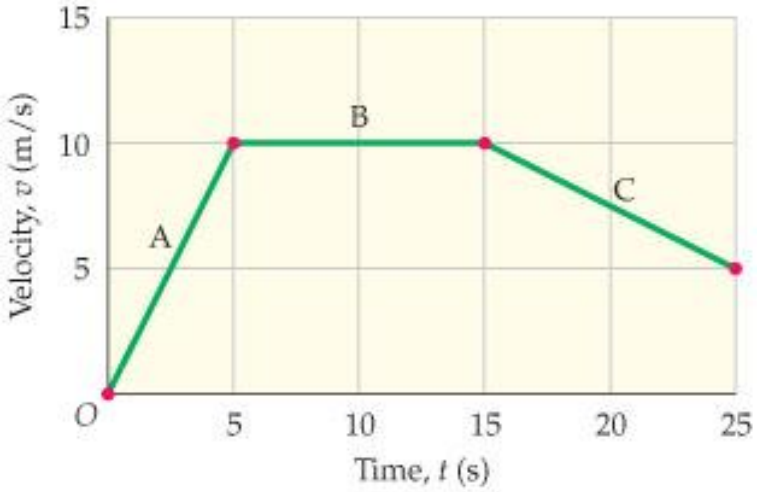
n	L_n	R_n
10	0.2850000	0.3850000
20	0.3087500	0.3587500
30	0.3168519	0.3501852
50	0.3234000	0.3434000
100	0.3283500	0.3383500
1000	0.3328335	0.3338335



- **Distances**

$$\text{distance} = \text{velocity} \times \text{time}$$

Example 5.10. An object is traveling along the x-axis in such a way that its velocity v at time t is shown in the following diagram . How far did it travel between $t = 0$s and $t = 25$s?



Exercises Page 375 5(a)(b) and find the area by using n rectangles and right endpoints with $n \to \infty$ (area = 6), 17.

5.2 The Definite Integral

- **Riemann Sums**

Consider a function defined on a closed interval $[a, b]$. Divide $[a, b]$ into n equal subintervals by means of points $a = x_0 < x_1 < x_2 < \ldots < x_{n-1} < x_n = b$, where

$$x_i = a + i\Delta x$$

and

$$\Delta x = \frac{b-a}{n}.$$

On each subinterval $[x_{i-1}, x_i]$, pick an arbitrary point x_i^*, which we call a sample point of the interval.

Let f be a function that is defined on the closed interval $[a, b]$. If

$$\lim_{n\to\infty} \sum_{i=1}^{n} f\left(x_i^*\right) \Delta x$$

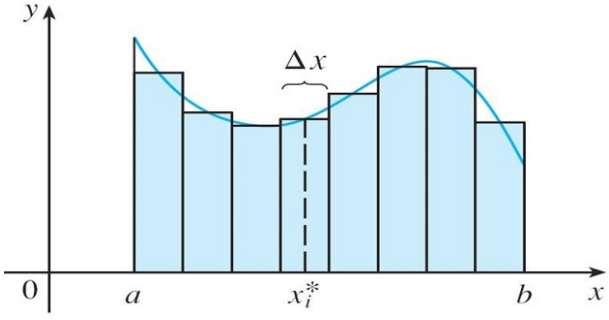
exists, we say that f is **integrable** on $[a, b]$, and denote this limit by $\int_a^b f(x)dx$, called the **definite integral** of f from a to b. Namely,

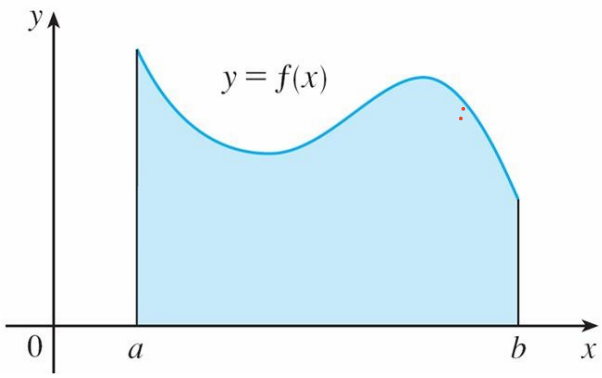
$$\int_a^b f(x)dx = \lim_{n\to\infty} \sum_{i=1}^{n} f\left(x_i^*\right) \Delta x$$

where $x_i^* = x_i = a + i\Delta x$

$\Delta x = \frac{b-a}{n}$

a and b are called the **lower limit** and **upper limit** of the integral, $f(x)$ is called the **integrand**.





Example 5.11. Evaluate $\int_{-1}^{3} (2x^2 - 8x)\,dx$ by using the definition of the integral.

$f(x)$

$$\Delta x = \frac{b-a}{n} = \frac{3+1}{n} = \frac{4}{n}$$

$$x_i^* = x_i = a + i\,\Delta x = -1 + i\,\frac{4}{n}$$

$$f(x_i^*)\Delta x = f\left(-1 + \frac{4}{n}i\right)\frac{4}{n}$$

$$= \left[2\left(-1+\frac{4}{n}i\right)^2 - 8\left(-1+\frac{4}{n}i\right)\right]\frac{4}{n}$$

$$= \frac{40}{n} - \frac{192}{n^2}i + \frac{128}{n^3}i^2$$

$$\sum_{i=1}^{n} f(x_i^*)\Delta x = \sum_{i=1}^{n}\left[\frac{40}{n} - \frac{192}{n^2}i + \frac{128}{n^3}i^2\right]$$

$$= \sum_{i=1}^{n}\frac{40}{n} - \sum_{i=1}^{n}\frac{192}{n^2}i + \sum_{i=1}^{n}\frac{128}{n^3}i^2$$

$$= \frac{40}{n}\sum_{i=1}^{n} 1 - \frac{192}{n^2}\sum_{i=1}^{n} i + \frac{128}{n^3}\sum_{i=1}^{n} i^2$$

$$\int_{-1}^{3}(2x^2 - 8x)dx =$$

$$= \frac{40}{n}(n) - \frac{192}{n^2}\left[\frac{n(n+1)}{2}\right] + \frac{128}{n^3}\left[\frac{n(n+1)(2n+1)}{6}\right]$$

$$= 40 - 96\left(1+\frac{1}{n}\right) + \frac{64}{3}\left(1+\frac{1}{n}\right)\left(2+\frac{1}{n}\right)$$

$$\int_{-1}^{3}(2x^2 - 8x)\,dx = \lim_{n\to\infty}\sum_{i=1}^{n} f(x_i^*)\Delta x$$

$$= \lim_{n\to\infty}\left[40 - 96\left(1+\frac{1}{n}\right) + \frac{64}{3}\left(1+\frac{1}{n}\right)\left(2+\frac{1}{n}\right)\right]$$

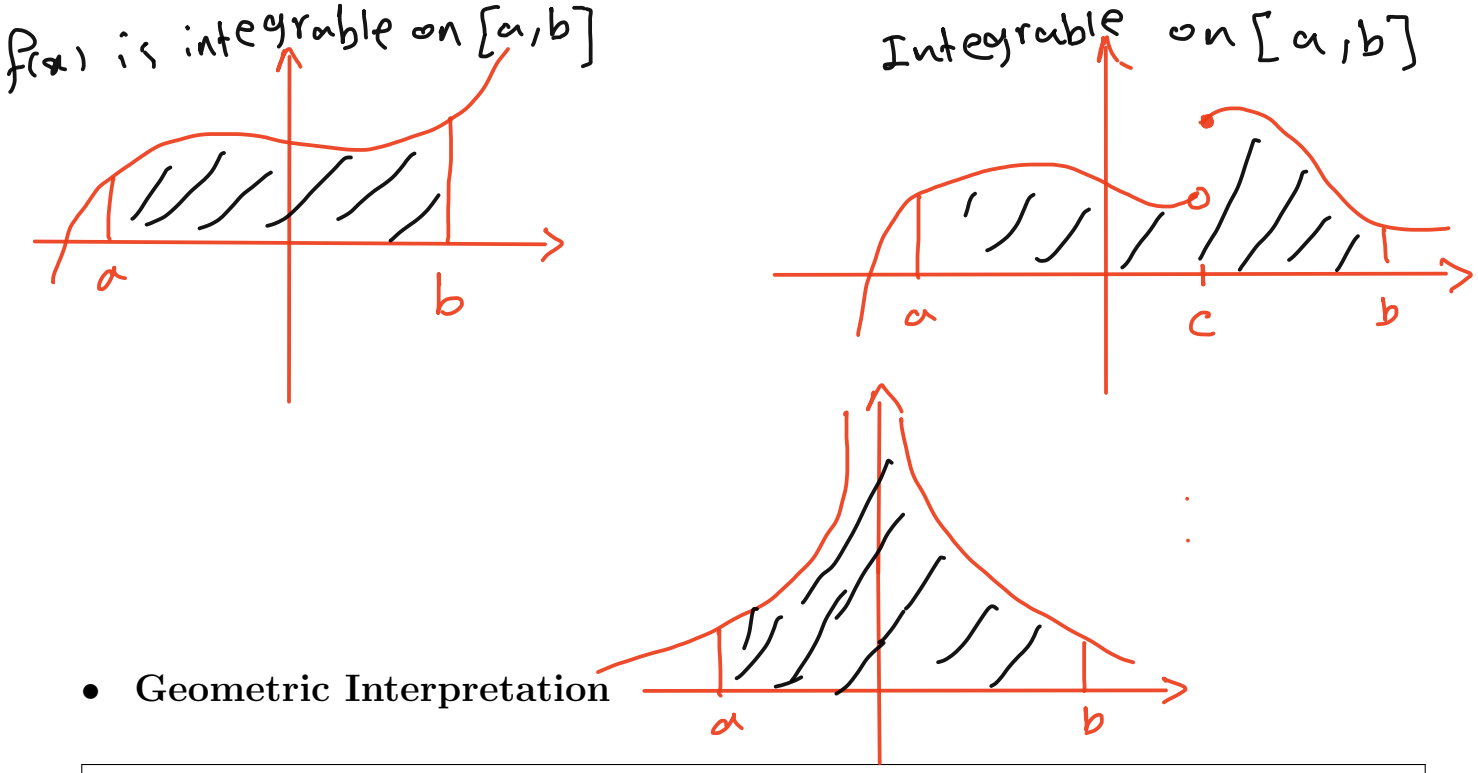
$$= 40 - 96 + \frac{128}{3} = \frac{-40}{3}$$

- **Theorem**

of jump discontinuous

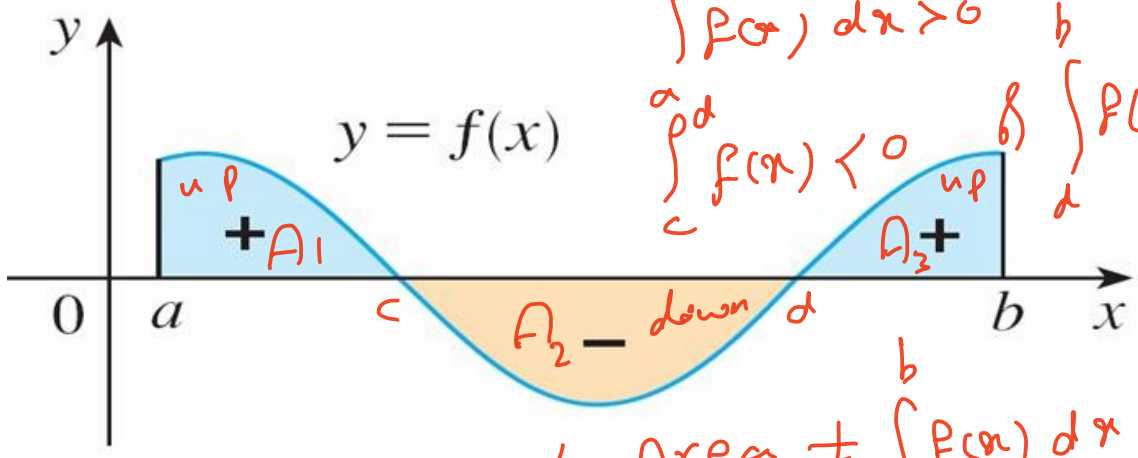
> If f is bounded on $[a,b]$ and if it is continuous there except at finite number ~~of points~~, then f is integrable on $[a,b]$.

> If f is continuous on $[a,b]$, then it is integrable on $[a,b]$.



- **Geometric Interpretation**

> $$\int_a^b f(x)dx = A_{\text{up}} - A_{\text{down}} = A_1 - A_2 + A_3$$



$\int_a^c f(x)\,dx > 0$

$\int_c^d f(x) < 0$

$\int_d^b f(x) > 0$

$\text{Area} = A_1 + A_2 + A_3$

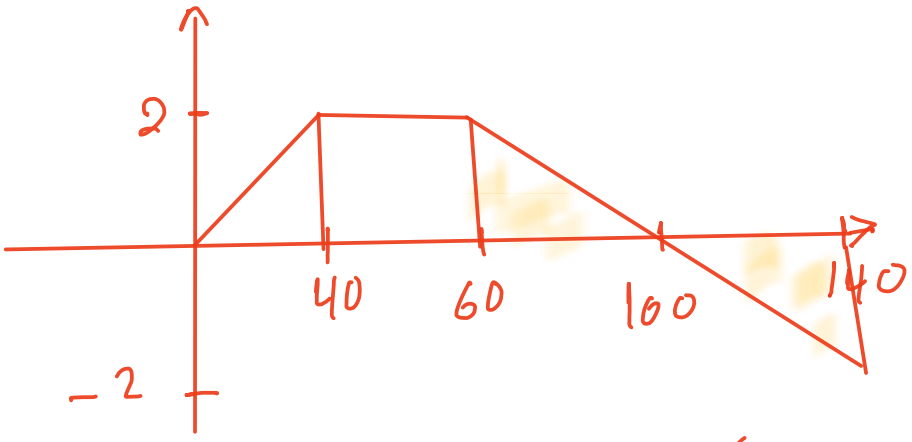
$\text{Area} \neq \int_a^b f(x)\,dx$

$\text{Area} = \int_a^c f(x)dx - \int_c^d f(x) + \int_a^b f(x)dx$

Example 5.12. An object at the origin at time $t = 0$ has velocity, measured in meters per second,

$$v(t) = \begin{cases} \dfrac{t}{20}, & 0 \le t \le 40 \\ 2, & 40 < t \le 60 \\ 5 - \dfrac{t}{20}, & t > 60 \end{cases}$$

Sketch the velocity curve. Express the position of the object at $t = 140$ s as a definite integral and evaluate it using formulas from plane geometry.



$$\int_0^{140} v(t)\,dt = \int_0^{40} v(t)\,dt + \int_{40}^{60} v(t)\,dt + \int_{60}^{100} v(t)\,dt + \int_{100}^{140} v(t)\,dt$$

$$= \int_0^{40} \frac{t}{20}\,dt + \int_{40}^{60} 2\,dt + \int_{60}^{100} 5 - \frac{t}{20}\,dt + \int_{100}^{140} 5 - \frac{t}{20}\,dt$$

Example 5.13. Evaluate $\int_0^3 \sqrt{9 - x^2}\,dx$.

$$= \frac{1}{2}(40)(2) + 2(20) + \frac{1}{2}(2)(40) - \frac{1}{2}(2)(40) = 80$$

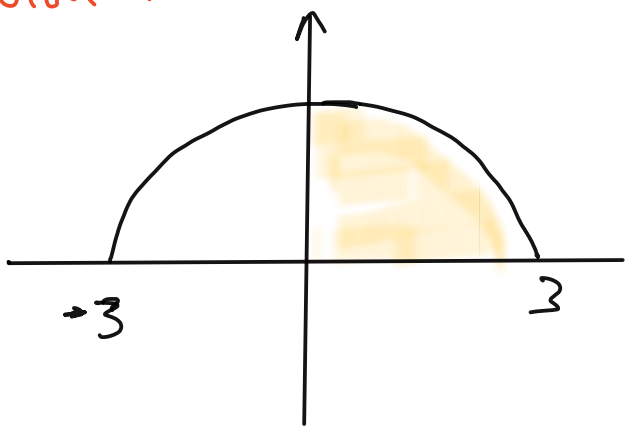
Ex 5.13 :o

$$\int_0^3 \sqrt{9-x^2}\,dx$$

$$y^2 + x^2 = 9$$

$$y^2 = 9 - x^2$$

$y = \sqrt{9-x^2}$ is semicircle with radius = 3



$$\int_0^3 \sqrt{9-x^2}\,dx = \frac{1}{4}\pi r^2$$

$$= \frac{1}{4}\pi(9)$$

$$= \frac{9}{4}\pi$$

- **Properties of the Definite Integral**

- $\int_a^a f(x)dx = 0.$
- $\int_a^c f(x)dx + \int_c^b f(x)dx = \int_a^b f(x)dx.$
- $\int_a^b f(x)dx = -\int_b^a f(x)dx.$

$\int_3^1 f(x) = -\int_1^3 f(x)\,dx$

Example 5.14. Given that $\int_0^{10} f(x)dx = 17$, $\int_0^8 f(x)dx = 12$, find $\int_8^{10} f(x)dx$.

$\int_8^{10} f(x)\,dx = ??$

$\int_8^{10} f(x)\,dx = \int_8^0 f(x)\,dx + \int_0^{10} f(x)$

$= -\int_0^8 f(x)\,dx + \int_0^{10} f(x)$

$= -12 + 17 = 5$

- **Linearity of the Definite Integral**

If f and g are integrable on $[a, b]$ and k is a constant, then kf and $f \pm g$ are integrable and

- $\int_a^b kf(x)dx = k\int_a^b f(x)dx$
- $\int_a^b [f(x) \pm g(x)]\,dx = \int_a^b f(x)dx \pm \int_a^b g(x)dx$

- **Comparison Property**

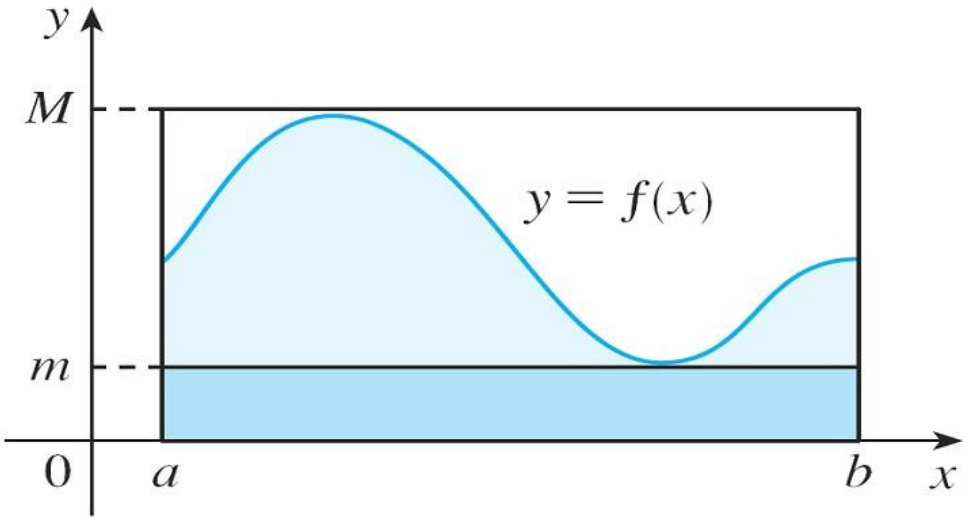
If f and g are integrable on $[a, b]$ and if $f(x) \leq g(x)$ for all $x \in [a, b]$, then

$$\int_a^b f(x)dx \leq \int_a^b g(x)dx$$

- **Boundedness Property**

If f is integrable on $[a,b]$ and $m \le f(x) \le M$ for all x in $[a,b]$, then

$$m(b-a) \le \int_a^b f(x)dx \le M(b-a)$$



Example 5.15. Give an estimate of $\int_0^1 e^{-x^2}dx$.

$0 \le x \le 1$

$0 \le x^2 \le 1$

$-1 \le -x^2 \le 0$

$e^{-1} \le e^{-x^2} \le e^0$

$\frac{1}{e} \le e^{-x^2} \le 1$

$\frac{1}{e}(1-0) \le \int_0^1 e^{-x^2}dx \le 1(1-0)$

$0.367 \le \int_0^1 e^{-x^2}dx \le 1$

Exercises Page 388 17, 19, 23, 35, 37, 39, 41, 47, 49, 55, 57.

5.3 The Fundamental Theorem of Calculus

- Calculus: Differential Calculus and Integral Calculus.
- Differential Calculus: arose from the tangent problem.
- Integral Calculus: arose from the area problem.
- Newton's mentor, Isaac Barrow ($1630 - 1677$), discovered that these two problems are actually closedly related.

HW

Example 5.16. Show that $\frac{d}{dx}\int_0^x t^2 dt = x^2$ using the definition of the integral.

$$\int_0^x t^2\,dx = \frac{x^3}{3}$$

By FFTC

$$\frac{d}{dx}\int_0^x t^2\,dt = x^2(1) - 0(0) = x^2$$

- **First Fundamental Theorem of Calculus**

Let f be continuous on the closed interval $[a, b]$ and let x be a variable point in (a, b). Then

$$\frac{d}{dx}\int_a^x f(t)dt = f(x)$$

Example 5.17. Find $\frac{d}{dx}\left[\int_1^x t^3 dt\right]$.

$$\frac{d}{dx}\int_1^x t^3\,dt = x^3(x)' - 1(1)'$$
$$= x^3$$

Example 5.18. Find $\frac{d}{dx}\left[\int_x^4 \tan^2 u \cos u\, du\right]$, $\frac{\pi}{2} < x < \frac{3\pi}{2}$.

$$\frac{d}{dx}\int_x^4 \tan^2 u \cos u\, du = \tan^2 4 \cos 4\,(0) - \tan^2 x \cos x\,(1)$$
$$= -\tan^2 x \cos x$$

Example 5.19. Find $D_x\left[\int_x^{x^2} \sqrt{3t^2-1}\,dt\right]$.

$$D_x\int_x^{x^2}\sqrt{3t^2-1}\,dt = \sqrt{3(x^2)^2-1}\,(2x) - \sqrt{3x^2-1}\,(1)$$
$$= 2x\sqrt{3x^4-1} - \sqrt{3x^2-1}$$

- **Second Fundamental Theorem of Calculus**

Let f be continuous on $[a, b]$, and let F be any antiderivative of f on $[a, b]$.

Then

$$\int_a^b f(x)dx = F(b) - F(a)$$

Example 5.20. Show that $\int_a^b c\,dx = c(b-a)$.

$$\int_a^b c\,dx = c\int_a^b x^0\,dx = c\left[\frac{x^{0+1}}{1}\right]_a^b = c\left[x\right]_a^b$$

$$= c\left[b-a\right] \quad \#$$

Example 5.21. Evaluate $\int_a^b x\,dx$.

$$= \left[\frac{x^2}{2}\right]_a^b = \frac{1}{2}\left[x^2\right]_a^b$$

$$= \frac{1}{2}\left[b^2 - a^2\right] = \frac{b^2 - a^2}{2}$$

Example 5.22. Given that r is a rational number, $r \neq -1$, evaluate $\int_a^b x^r\,dx$.

$$\int_a^b x^r\,dx = \left[\frac{x^{r+1}}{r+1}\right]_a^b = \frac{1}{r+1}\left[x^{r+1}\right]_a^b = \frac{1}{r+1}\left[b^{r+1} - a^{r+1}\right]$$

Example 5.23. Evaluate $\int_{-1}^{2} (4x - 6x^2)\,dx$.

$$\int_{-1}^{2} (4x - 6x^2)\,dx = \left[4\frac{x^2}{2} - 6\frac{x^3}{3}\right]_{-1}^{2} = 2\left[x^2 - x^3\right]_{-1}^{2}$$

$$= 2\left[(4 - 8) - (1 + 1)\right] = 2\left[-4 - 2\right] = -12$$

Example 5.24. Evaluate $\int_1^8 \left(x^{\frac{1}{3}} + x^{\frac{4}{3}}\right) dx$.

$$\int_1^8 \left(x^{\frac{1}{3}} + x^{\frac{4}{3}}\right) dx = \left[\frac{3}{4}x^{\frac{4}{3}} + \frac{3}{7}x^{\frac{7}{3}}\right]_1^8$$

$$= \left(\frac{3}{4}\sqrt[3]{8^4} + \frac{3}{7}\sqrt[3]{8^7}\right) - \left(\frac{3}{4} + \frac{3}{7}\right) = \frac{1839}{28}$$

Example 5.25. Find $\int_0^x 3\sin t dt$.

$$\int_0^x 3\sin t\, dt = -3\left[\cos t\right]_0^x = -3(\cos x - \cos 0) = -3(\cos x - 1)$$

Example 5.26. Evaluate $\int_1^3 e^x dx$.

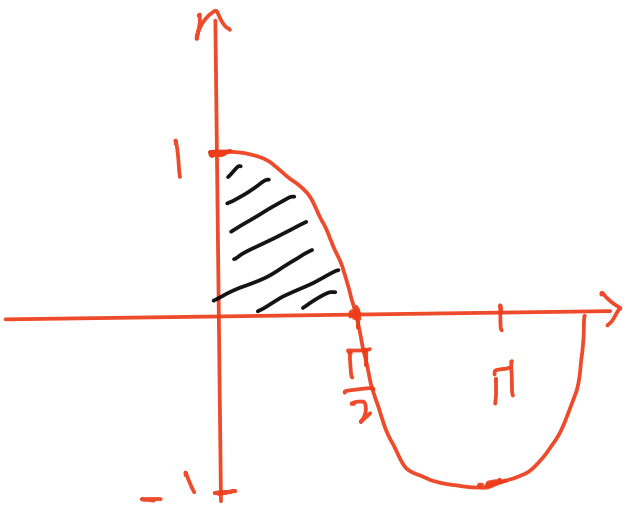
$$\int_1^3 e^x dx = \left[e^x\right]_1^3 = e^3 - e$$

Example 5.27. Evaluate $\int_3^6 \frac{1}{x} dx$.

$$\int_3^6 \frac{1}{x}\,dx = \Big[\ln|x|\Big]_3^6 = \ln|6| - \ln|3| = \ln 6 - \ln 3$$

$$= \ln\frac{6}{3} = \ln 2$$

Example 5.28. Find the area under the curve $y = \cos x$ from $x = 0$ to $x = \frac{\pi}{2}$.



$$\int_0^{\frac{\pi}{2}} f(x)\,dx = A_{up}$$

$$\text{Area} = \int_0^{\frac{\pi}{2}} \cos x\,dx$$

$$= \Big[\sin x\Big]_0^{\frac{\pi}{2}}$$

$$= \sin\frac{\pi}{2} - \sin 0 = 1 - 0 = 1$$

Exercises Page 399 7, 11, 13, 17, 21, 25, 29, 31, 37, 41, 43, 47, 61, 67, 75.

5.4 Indefinite Integrals and the Net Change Theorem

- The notation $\int f(x)dx$ is traditionally used for an antiderivative of f and is called an **indefinite integral**.

$$\int f(x)dx = F(x) + C \quad \text{means} \quad F'(x) = f(x)$$

- **Standard Integral Formulas**

1. $\int kdu = ku + C$
2. $\int u^r du = \dfrac{u^{r+1}}{r+1} + C \quad (r \neq -1)$
3. $\int \dfrac{1}{u}du = \ln|u| + C$
4. $\int e^u du = e^u + C$
5. $\int a^u du = \dfrac{a^u}{\ln a} + C \quad (a > 0,\ a \neq -1)$
6. $\int \sin u du = -\cos u + C$
7. $\int \cos u du = \sin u + C$
8. $\int \sec^2 u du = \tan u + C$
9. $\int \csc^2 u du = -\cot u + C$
10. $\int \sec u \tan u du = \sec u + C$
11. $\int \csc u \cot u du = -\csc u + C$
12. $\int \dfrac{1}{\sqrt{1-u^2}}du = \sin^{-1} u + C$
13. $\int \dfrac{1}{1+u^2}du = \tan^{-1} u + C$
14. $\int \sinh u du = \cosh u + C$
15. $\int \cosh u du = \sinh u + C$

14. $\int \sec x\,dx = \ln|\sec x + \tan x| + C$

Example 5.29. Evaluate $\int (10x^4 - 2\sec^2 x)\,dx$.

$\int (10x^4 - 2\sec^2 x)\,dx = 10\cdot\frac{x^5}{5} - 2\tan x + C$

$= 2x^5 - 2\tan x + C$

Example 5.30. Evaluate $\int \frac{\cos\theta}{\sin^2\theta}d\theta$. $= \int \frac{\cos\theta}{\sin\theta}\cdot\frac{1}{\sin\theta}\,d\theta$

$= \int \cot\theta \csc\theta\,d\theta = -\csc\theta + C$

Example 5.31. Evaluate $\int_0^3 (x^3 - 6x)\,dx$. $= \frac{-27}{4}$ #W

Example 5.32. Evaluate $\int_0^2 \left(2x^3 - 6x + \frac{3}{x^2+1}\right) dx.$

$$2\int_0^2 x^3\,dx - 6\int_0^2 x\,dx + 3\int_0^2 \frac{1}{x^2+1}\,dx$$

$$= \left[\frac{2}{4}x^4 - \frac{6}{2}x^2 + 3\tan^{-1}x\right]_0^2 = (8 - 12 + 3\tan^{-1}2) - (0 - 0 + 0)$$

$$= -4 + 3\tan^{-1}2$$

Example 5.33. Evaluate $\int_1^9 \frac{2t^2 + t^2\sqrt{t} - 1}{t^2} dt.$

$$\int_1^9 \left(\frac{2t^2}{t^2} + \frac{t^2\sqrt{t}}{t^2} - \frac{1}{t^2}\right) dt = \int_1^9 \left(2 + t^{\frac{1}{2}} - t^{-2}\right) dt$$

$$= \left[2t + \frac{2}{3}t^{\frac{3}{2}} + \frac{1}{t}\right]_1^9 = \left(18 + \frac{54}{3} + \frac{1}{9}\right) - \left(2 + \frac{2}{3} + 1\right)$$

$$= \frac{292}{9}$$

- **Distance and Displacement**

Consider an object travelling along a straight line with velocity $v(t)$ at time t.

(a) $\int_a^b v(t)dt$ measures the **displacement** (*net change of position*) of the object in the time interval $a \leq t \leq b$.

(b) $\int_a^b |v(t)|dt$ measures the **distance travelled** by the object from time $t = a$ to $t = b$.

Example 5.34. A particle moves along a line so that its velocity at time t is $v(t) = (t^2 - t - 6)$ m/s.

(a) Find the displacement of the particle during the time period $1 \leq t \leq 4$.

(b) Find the distance traveled during this time period.

Exercises Page 408 1, 3, 9, 13, 17, 23, 25, 27, 29, 33, 37, 41, 45, 59, 61.

5.5 The Substitution Rule

- **Substitution Rule for Indefinite Integrals**

> Let g be a differentiable function and suppose that F is an antiderivative of f, then
>
> $$\int f(g(x))\, g'(x)dx = F(g(x)) + C$$

Example 5.35. Evaluate $\int x^3\sqrt{x^4+11}dx$. $= \int x^3 (x^4+11)^{\frac{1}{2}} dx$

let $u = x^4 + 11$

$\frac{du}{dx} = 4x^3 \iff du = 4x^3 dx \iff \div 4 \iff \frac{1}{4} du = x^3 dx$

$\int x^3 (x^4+11)^{\frac{1}{2}} dx = \int \frac{1}{4} u^{\frac{1}{2}} du = \frac{1}{4}\int u^{\frac{1}{2}} du$

$= \frac{1}{4} \frac{u^{\frac{3}{2}}}{\frac{3}{2}} + C = \frac{1}{4}\left(\frac{2}{3}\right) u^{\frac{3}{2}} + C$

$= \frac{1}{6} (x^4+11)^{\frac{3}{2}} + C$

Example 5.36. Evaluate $\int \sqrt{x^2+x}(2x+1)dx$.

$$\int \sqrt{x^2+x}\,(2x+1)\,dx$$

let $u = x^2 + x$

$$\frac{du}{dx} = 2x+1 \iff du = (2x+1)\,dx$$

$$\int \sqrt{x^2+x}\,(2x+1)dx = \int u^{\frac{1}{2}}\,du = \frac{2}{3}u^{\frac{3}{2}} + C$$

$$= \frac{2}{3}(x^2+x)^{\frac{3}{2}} + C$$

Example 5.37. Evaluate $\int \frac{x^5}{\sqrt{x^2+1}}dx$.

let $u = x^2+1 \iff x^2 = u-1 \iff x^4 = (u-1)^2$

$$\frac{du}{dx} = 2x \iff du = 2x\,dx \iff \frac{1}{2}du = x\,dx$$

$$\int \frac{x^5\,dx}{\sqrt{x^2+1}} = \int \frac{x\,x^4}{\sqrt{x^2+1}}dx = \int \frac{\frac{1}{2}(u-1)^2}{u^{\frac{1}{2}}}du$$

$$= \frac{1}{2}\int \frac{u^2-2u+1}{u^{\frac{1}{2}}}du = \frac{1}{2}\int\left(u^{\frac{3}{2}} - 2u^{\frac{1}{2}} + u^{-\frac{1}{2}}\right)du$$

$$= \frac{1}{2}\left[\frac{2}{5}u^{\frac{5}{2}} - 2\left(\frac{2}{3}\right)u^{\frac{3}{2}} + 2u^{\frac{1}{2}}\right] + C$$

$$= \frac{1}{5}(x^2+1)^{\frac{5}{2}} - \frac{2}{3}(x^2+1)^{\frac{3}{2}} + (x^2+1)^{\frac{1}{2}} + C$$

Example 5.38. Find $\int \sin 3x dx$.

let $u = 3x$

$\frac{du}{dx} = 3 \iff du = 3\,dx \iff \frac{1}{3}du = dx$

$\int \sin 3x\,dx = \int \frac{1}{3}\sin u\,du = \frac{1}{3}\int \sin u\,du$

$= \frac{-1}{3}\cos u + C = \frac{-1}{3}\cos 3x + C$

Example 5.39. Evaluate $\int x \sin x^2 dx$.

let $u = x^2$

$\frac{du}{dx} = 2x \iff x\,dx = \frac{1}{2}du$

$\int x \sin x^2\,dx = \frac{1}{2}\int \sin u\,du = \frac{-1}{2}\cos u + C = \frac{-1}{2}\cos x^2 + C$

Example 5.40. Evaluate $\int \sin^3 2x \cos 2x dx$.

let $u = \sin 2x$

$\frac{du}{dx} = 2\cos 2x \iff \cos 2x\,dx = \frac{1}{2}du$

$\int \sin^3 2x \cos 2x\,dx = \frac{1}{2}\int u^3\,du = \frac{1}{2}\left(\frac{1}{4}\right)u^4 + C$

$= \frac{1}{8}\sin^4 2x + C$

Example 5.41. Evaluate $\int \tan x dx$.

$$\int \tan x \, dx = \int \frac{\sin x}{\cos x} dx$$

let $u = \cos x$

$$\frac{du}{dx} = -\sin x \cdot \Leftrightarrow \sin x \, dx = -du$$

$$\int \frac{\sin x}{\cos x} dx = -\int \frac{du}{u} = -\ln|u| + C = -\ln|\cos x| + C$$

$$= \ln\left|\frac{1}{\cos x}\right| + C = \ln|\sec x| + C$$

Example 5.42. Evaluate $\int \sec x dx$.

$$\int \sec x \, dx = \int \sec x \times \frac{\sec x + \tan x}{\sec x + \tan x} dx$$

$$= \int \frac{\sec^2 x + \sec x \tan x}{\sec x + \tan x} dx$$

let $u = \sec x + \tan x$

$$\frac{du}{dx} = \sec x \tan x + \sec^2 x \Leftrightarrow (\sec x \tan x + \sec^2 x) dx = du$$

$$= \int \frac{du}{u} = \ln|u| + C = \ln|\sec x + \tan x| + C$$

- **Substitution Rule for Definite Integrals**

> Let g have continuous derivative on $[a, b]$, and let f be continuous on the range of g, then
>
> $$\int_a^b f\left(g(x)\right) g'(x)dx = \int_{g(a)}^{g(b)} f(u)du$$
>
> where $u = g(x)$.

Example 5.43. Evaluate $\int_0^4 \sqrt{2x+1}dx$.

let $u = 2x+1$

$\frac{du}{dx} = 2 \iff dx = \frac{1}{2}du$

when $x=0$, $u=1$

when $x=4$, $u=9$

$$\frac{1}{2}\int_1^9 \sqrt{u}\,du = \frac{1}{2}\int_1^9 u^{\frac{1}{2}}du$$

$$= \frac{1}{2}\left(\frac{2}{3}\right)\left[u^{\frac{3}{2}}\right]_1^9$$

$$= \frac{1}{3}\left[27-1\right] = \frac{26}{3}$$

HW

Example 5.44. Evaluate $\int_0^1 \frac{x+1}{(x^2+2x+6)^2}dx$. $= \frac{1}{36}$

H W

Example 5.45. Evaluate $\int_{\frac{\pi^2}{9}}^{\frac{\pi^2}{4}} \frac{\cos\sqrt{x}}{\sqrt{x}}dx.$ $= 2 - \sqrt{3}$

H W

Example 5.46. Evaluate $\int_1^e \frac{\ln x}{x}dx.$ $= \frac{1}{2}$

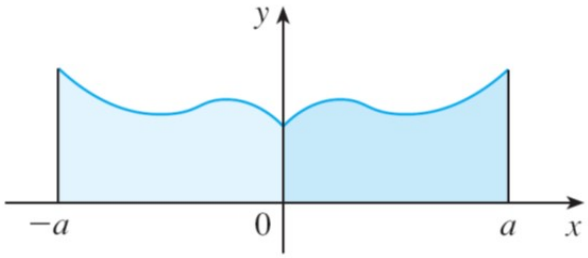
- **Symmetry Theorem**

If f is an even function, then

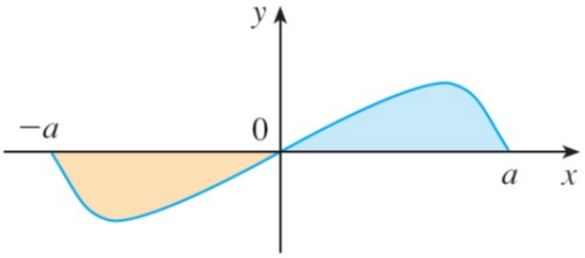
$$\int_{-a}^{a} f(x)dx = 2\int_{0}^{a} f(x)dx$$

If f is an odd function, then

$$\int_{-a}^{a} f(x)dx = 0$$



(a) f even, $\int_{-a}^{a} f(x)\,dx = 2\int_{0}^{a} f(x)\,dx$



(b) f odd, $\int_{-a}^{a} f(x)\,dx = 0$

Example 5.47. Evaluate $\int_{-\pi}^{\pi} \cos\left(\frac{x}{4}\right) dx$.

Example 5.48. Evaluate $\int_{-5}^{5} \frac{x^5}{x^2+4} dx$.

Example 5.49. Evaluate $\int_{-2}^{2} \left(x \sin^4 x + x^3 - x^4\right) dx$.

Example 5.50. Evaluate $\int_{-\pi}^{\pi} \sin^3 x \cos^5 x dx$.

Exercises Page 418 7, 9, 11, 13, 15, 17, 19, 21, 23, 25, 27, 33, 35, 39, 41, 45, 47, 57, 59, 61, 65, 67, 69, 71, 73, 77, 81, 87, 91, 93.